

Cartilage and Bone Histology

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Cartilage and Bone

- Dense connective tissue includes:
 - Dense irregular connective tissue
 - Tendons
 - Ligaments
 - Cartilage
 - Bone

Information on Connective Tissues

- Function is to bear large loads
- These tissues consist of cells that produce a matrix

cell is one entity and spits outs all around it it's ec matrix

Cartilage

- Cartilage is a specialized connective tissue composed of:
 - Cells called chondrocytes that live in cavities called lacunae
 - Extra cellular matrix = free and fixed glycosaminoglycans (GAGs), proteoglycans, collagen

outside of the lacunae

Unique Properties of Cartilage

- NO direct nerve, blood, or lymph supply
- Chondrocytes have a low metabolic rate
- High tensile strength
- Permits reversible compression

Function of Cartilage

- Support
- Friction free surface at articulations
- Participates in load distribution
- Precursor for development and growth of bones

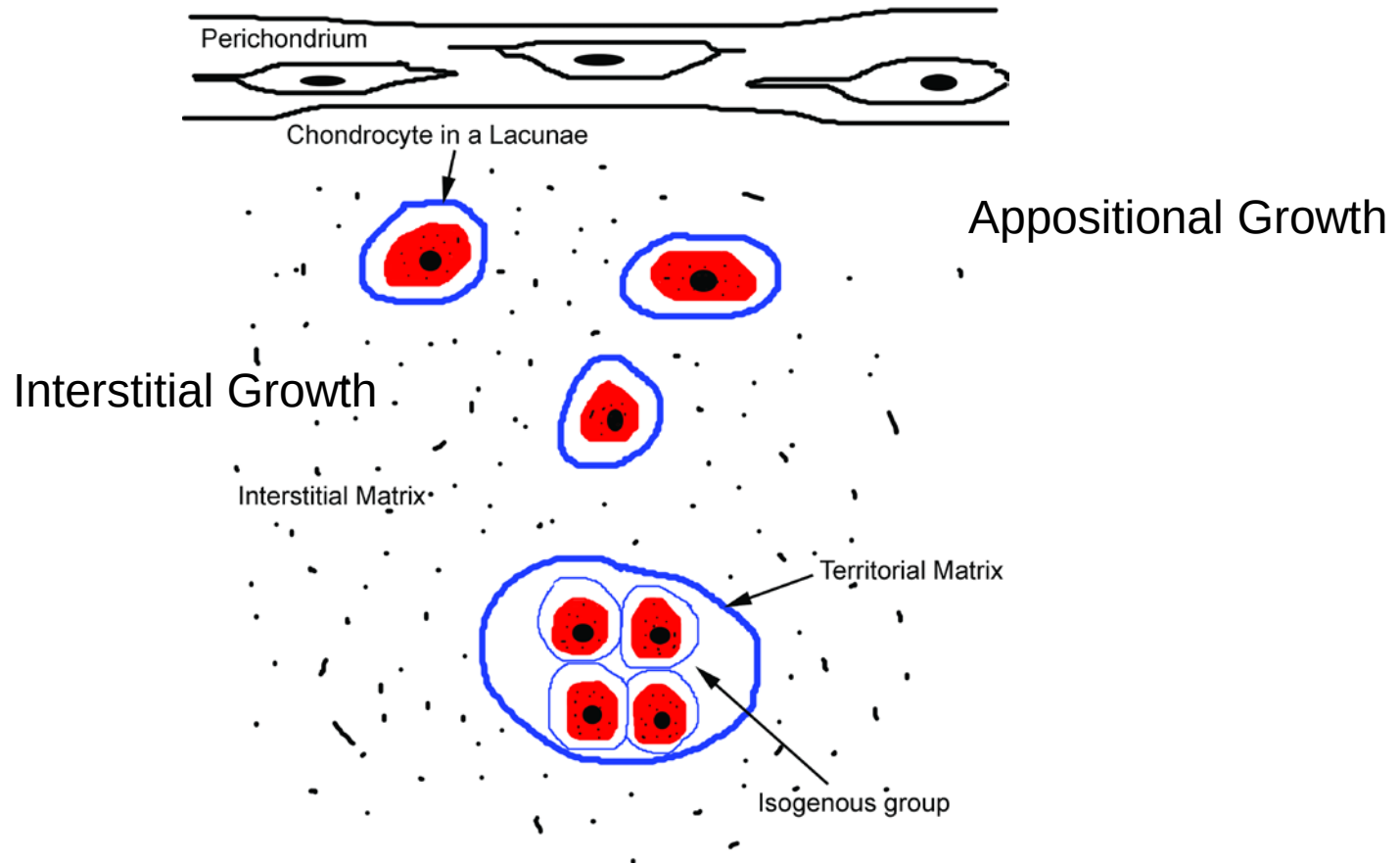
Cartilage and Perichondrium

- Some types of cartilage have a covering called perichondrium
- Perichondrium is a dense irregular connective tissue that serves as an interface between cartilage and neighboring tissue
- Perichondrium is highly vascular and provides nutrition to chondrocytes via diffusion
- There are 2 histological layers to perichondrium: Outer fibrous layer and inner cellular layer
- The inner cellular layer can differentiate into chondroblasts which will make a matrix and become chondrocytes

Growth of Cartilage

- There are 2 mechanisms for cartilage growth:
 1. Interstitial growth = mitotic division of chondrocytes form isogenous groups
 2. Appositional growth = inner layer of the perichondrium has cells that can differentiate into chondroblasts that create a matrix and become chondrocytes

Growth of Cartilage



Types of Cartilage

1. Hyaline cartilage
2. Elastic cartilage
3. Fibrocartilage

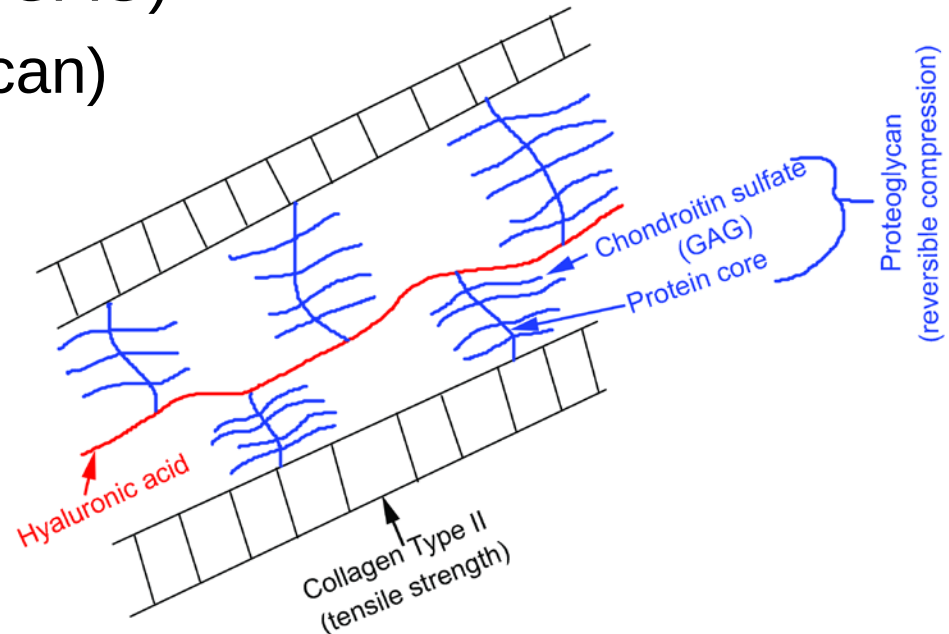
Hyaline Cartilage

- Found in:
 - Articular surfaces of bones
 - Walls of respiratory passages
 - Anterior ends of ribs
 - Epiphyseal plates of long bones
- Appearance
 - Glassy, translucent, bluish/white

Composition of Hyaline Cartilage

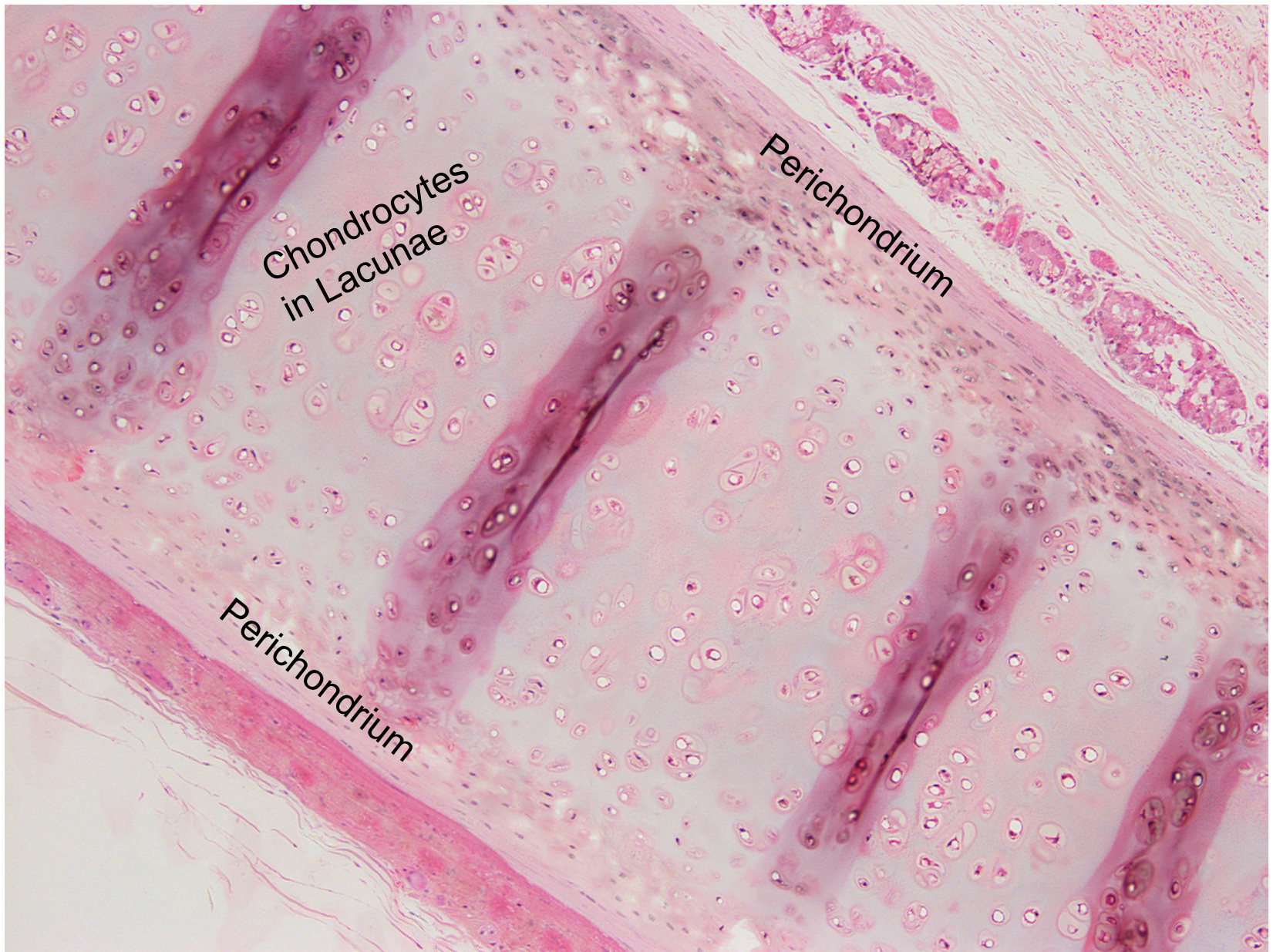
- Chondrocytes in lacunae
- Matrix consists of
 - Type II collagen fibrils
 - Matrix proteins
 - Hyaluronic acid (free GAG)
 - Aggrecan (Proteoglycan)

Image to the right depicts
hyaline cartilage matrix



Development of Hyaline Cartilage

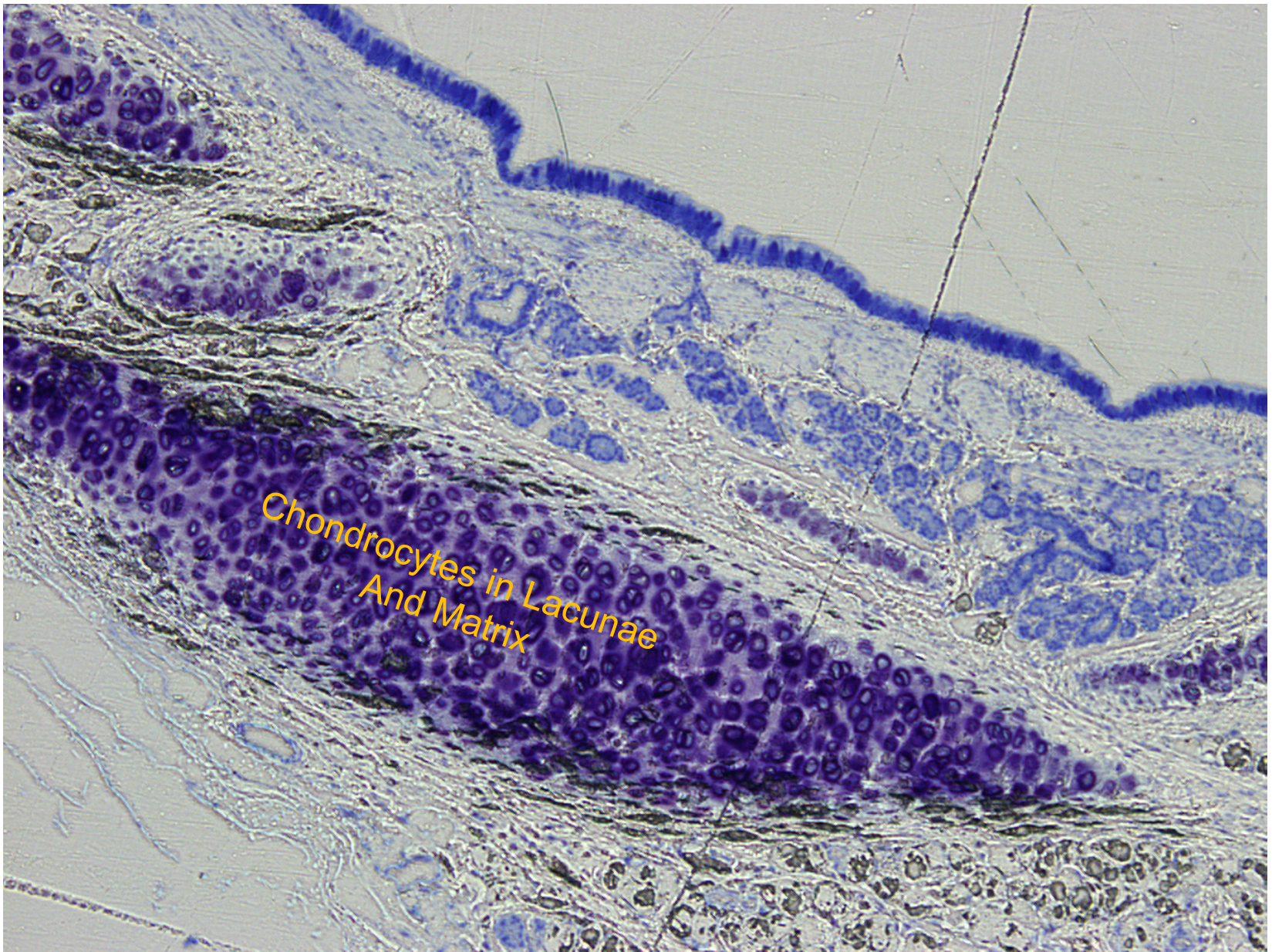
- Mesenchymal cells at the periphery form fibroblasts which make dense irregular connective tissue called perichondrium.
- The inner layer of the perichondrium can differentiate into chondroblasts which then create a matrix and become chondrocytes. This type of growth is called appositional growth.
- Chondrocytes that go through mitosis and form isogenous groups = interstitial growth



Trachea and esophagus – hyaline cartilage



Trachea and esophagus – hyaline cartilage (high magnification)



Chondrocytes in Lacunae
And Matrix

Lungs and bronchi – hyaline cartilage

Elastic Cartilage

- Found in regions that require flexible support and reversible compression
 - Auricle
 - Eustacian tube
 - Epiglottis
 - Laryngeal cartilages
- The extra cellular matrix consists of elastic fibers in addition to the other constituents

Elastic Cartilage

Fibrous layer of perichondrium



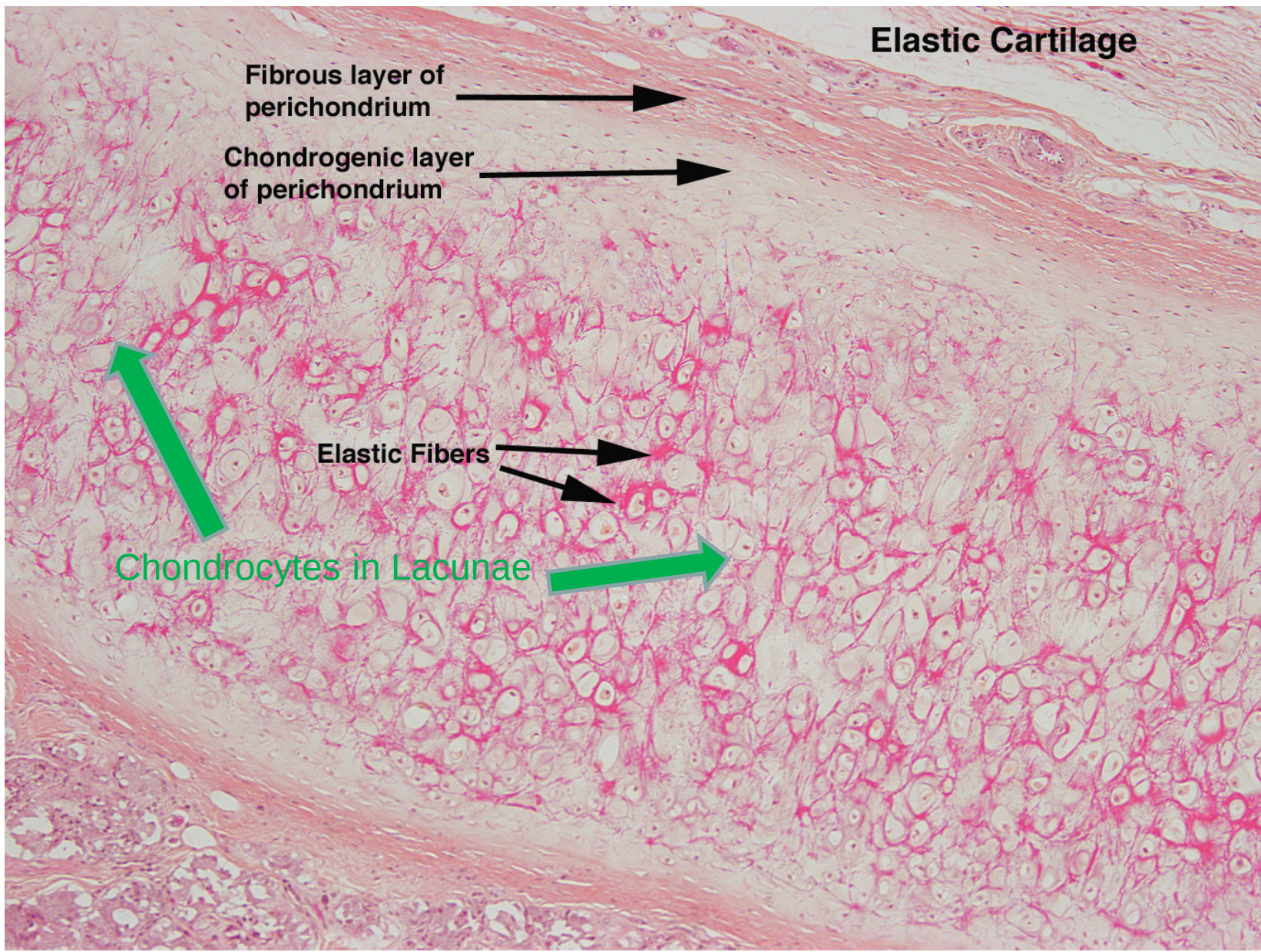
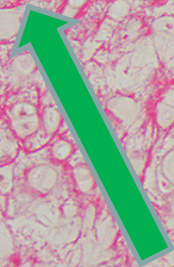
Chondrogenic layer of perichondrium



Elastic Fibers

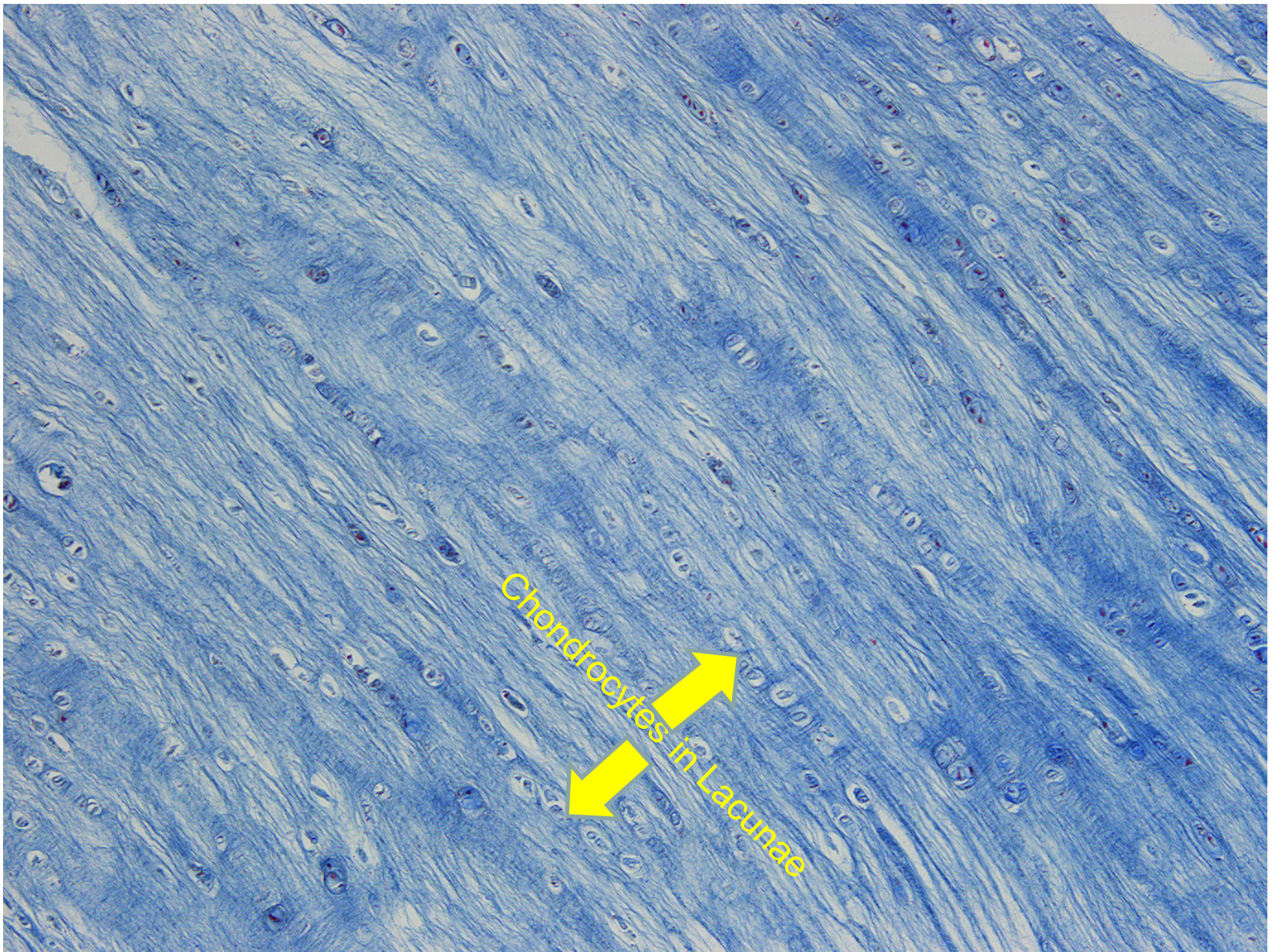


Chondrocytes in Lacunae



Fibrocartilage

- Found in regions needing firm support and tensile strength
 - Intervertebral disks
 - Pubic symphysis
 - Attachments of tendons and ligaments to bone
- Is a transitional tissue that merges tendons, ligaments or hyaline cartilage
- Unique features
 - NO Perichondrium
 - Chondrocytes are in linear isogenous groups
- Development: Mesenchyme develops into chondroblasts



Fibrocartilage

Bone

- Specialized connective tissue with a calcified matrix

Properties

- Highly vascular
- Nutrition via cell to cell contact through canaliculi
- Appositional growth only
- Strong in tension, weak in torsion

Function

- Support/protection
- Storage space for calcium phosphate
- Transmits force of muscle contraction for locomotion

Bone cells

- Osteoprogenitor cells are derived from 2 separate cell lines:
 1. Mesenchymal stem cells produce osteoblasts
 2. Monocyte/macrophage line produces osteoclasts

Osteoblast

large cell, secretes matrix

Osteocyte

an osteoblast trapped in matrix

Osteoclast

large multinucleated cell that resides in a "Howships"
lacuna
resorbs bone

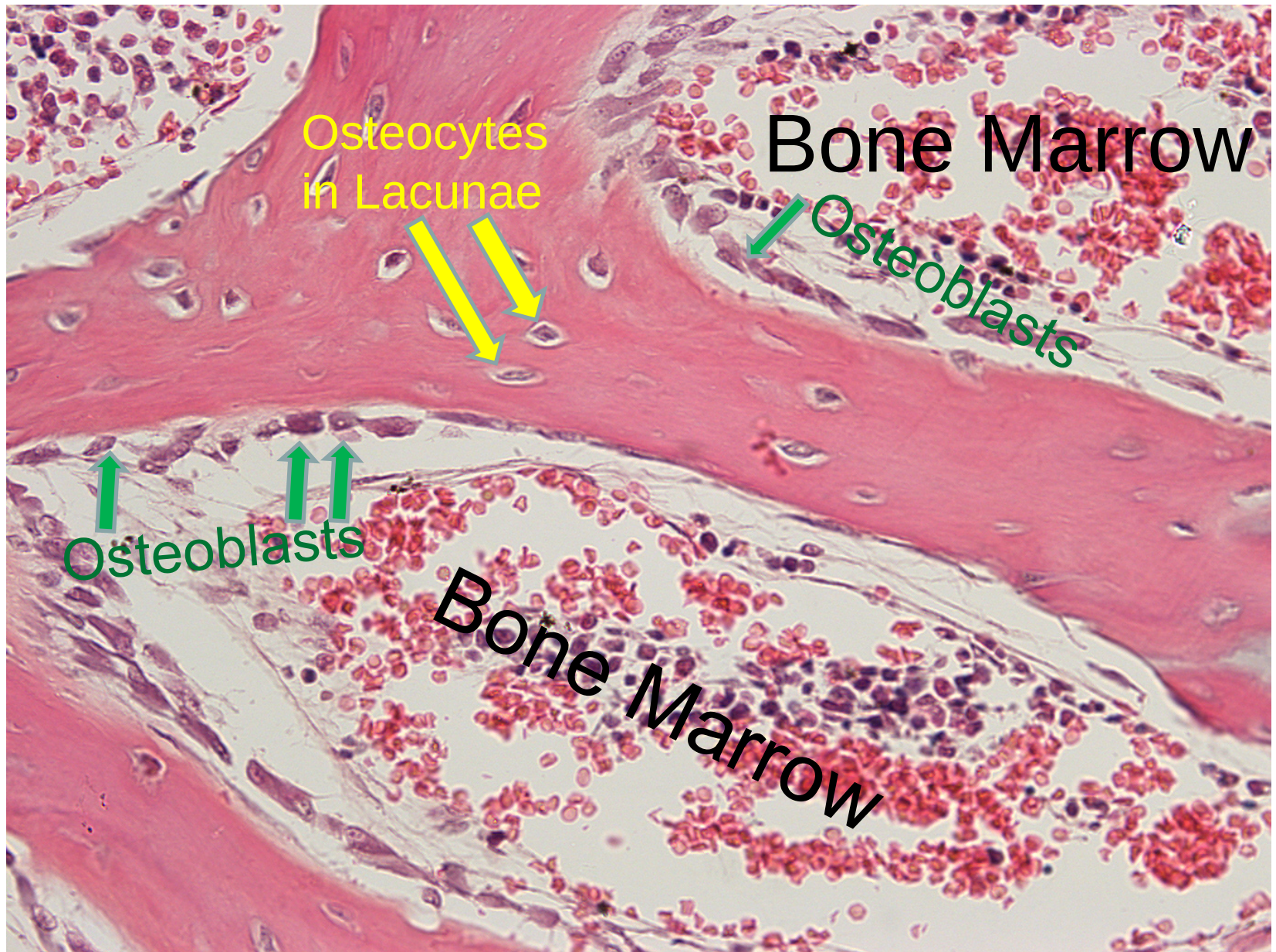
Regions of Bone

1. Trabecular (cancellous/spongy) bone

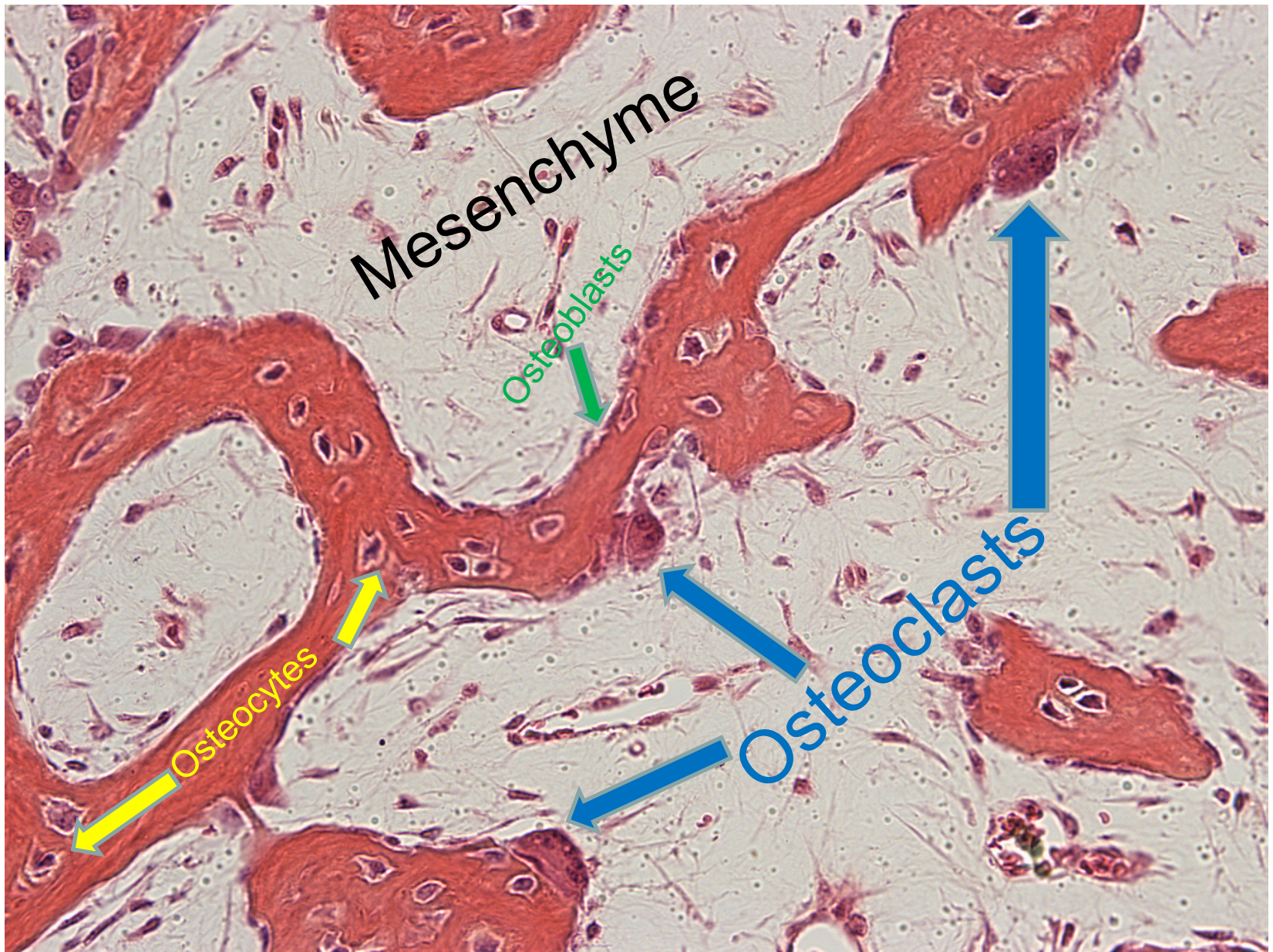
- Lattice or fine strips of bone that surrounds cavities of bone marrow

2. Cortical bone (compact bone) solid bone

Femur is 80% compact bone and 20% trabecular bone



Developing bone



Developing bone - head

Bone lining layers

1. Periosteum

Consists of 2 layers

- Outer layer = connective tissue layer (fibroblasts & collagen)
- Inner layer = osteogenic layer

Function

Supplies osteogenic cells for bone growth

Nutrition for osteocytes

Sharpey's fibers = bundles of collagen that anchor periosteum to bone

2. Endosteum

Single layer of cells that lines all inner bone surfaces

Contains osteoblasts and osteoprogenitor cells

Function

Nutrition, bone growth, repair

Microscopic Structure of Bone

1. Cortical Bone

Osteocytes in lacuna are in layers = lamellae

Numerous canaliculi – projections between osteocytes

There are 3 types of lamellae;

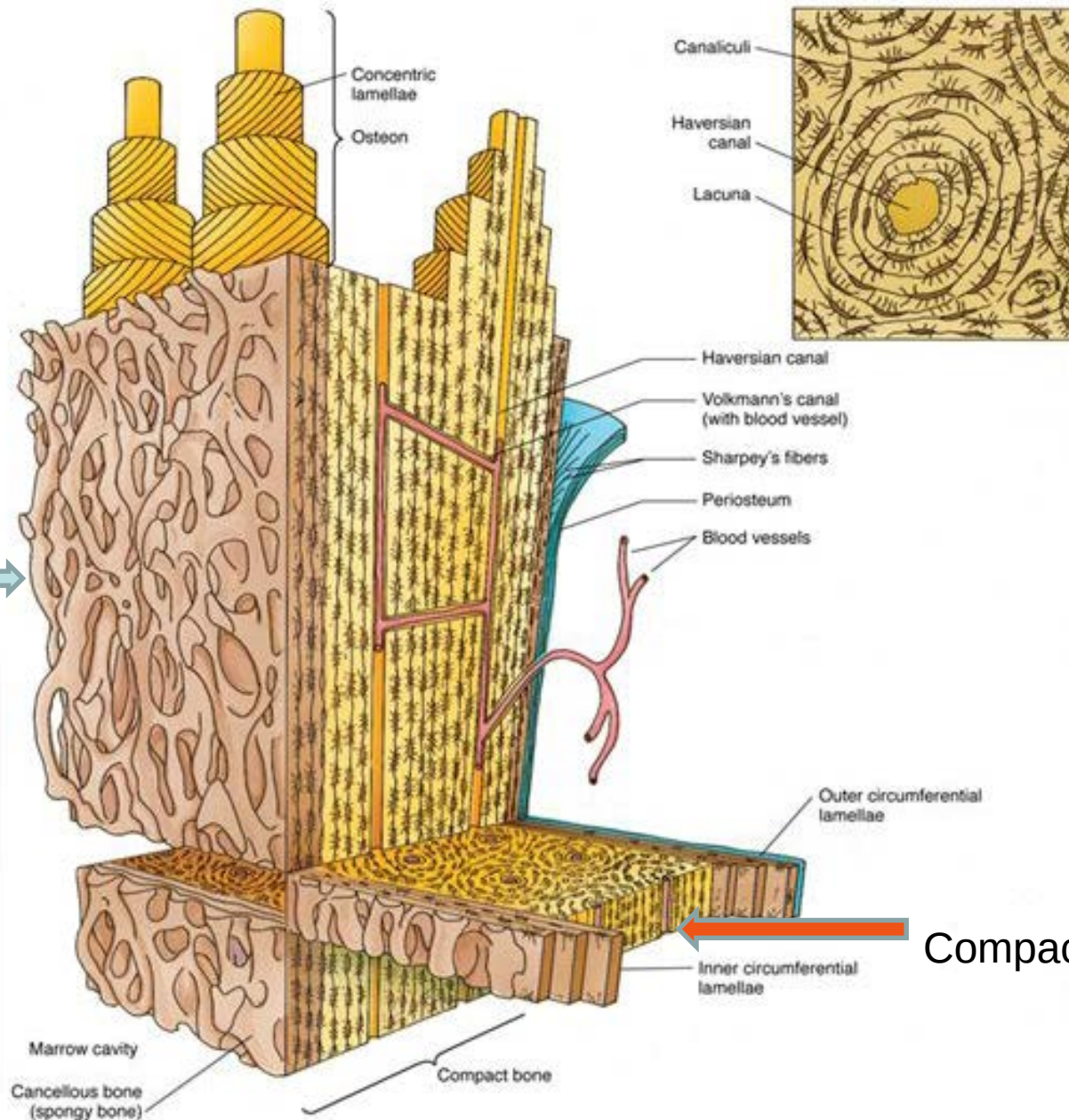
1. Concentric lamellae = Haversian system = osteon
cylindrical arrangement of compact bone
center of the osteon contains the
Haversian canal

2. Circumferential lamellae

3. Interstitial lamellae

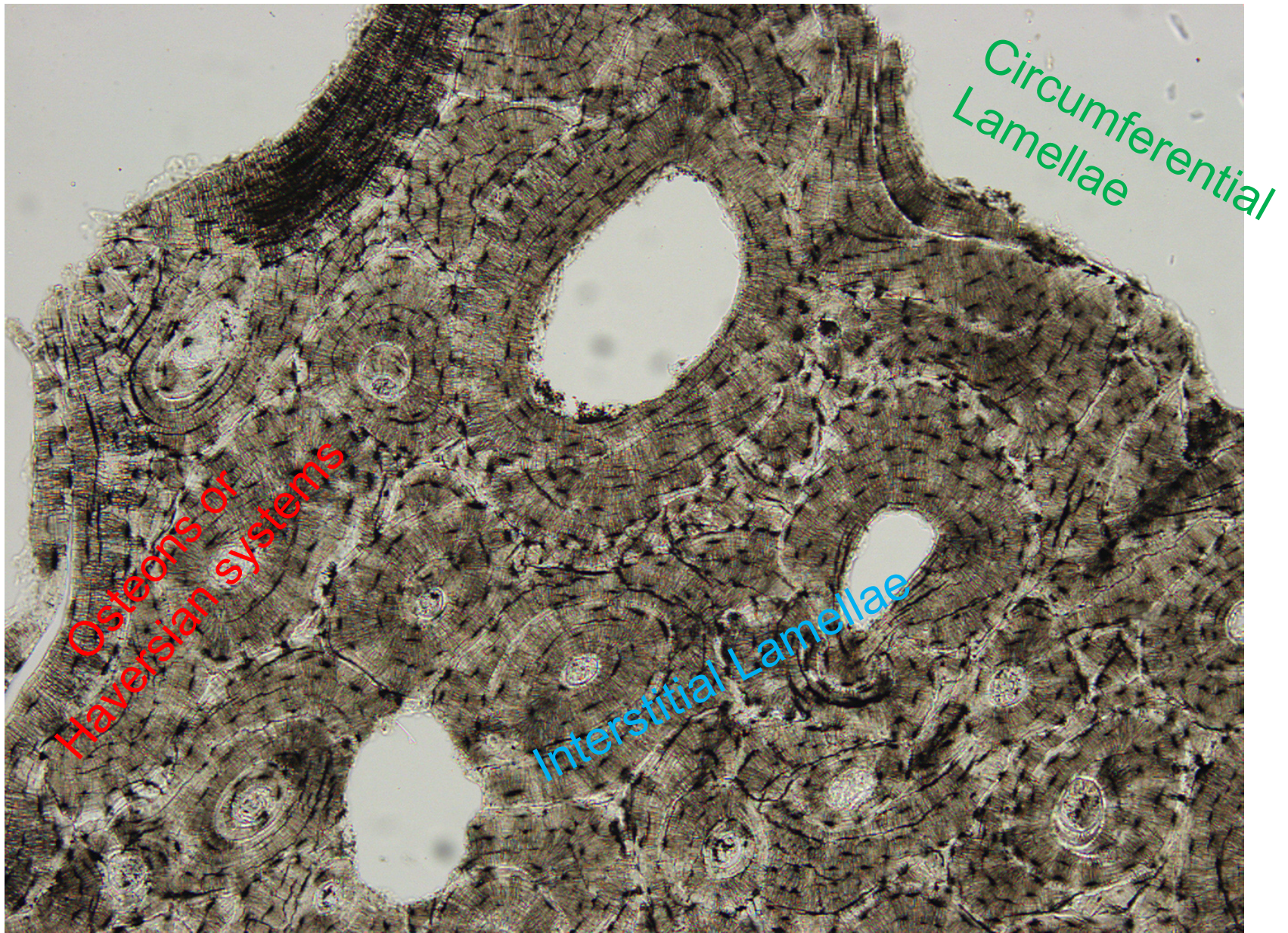
Is a partially eroded osteon which indicates bone remodeling

Trabecular/Cancellous/Spongy bone

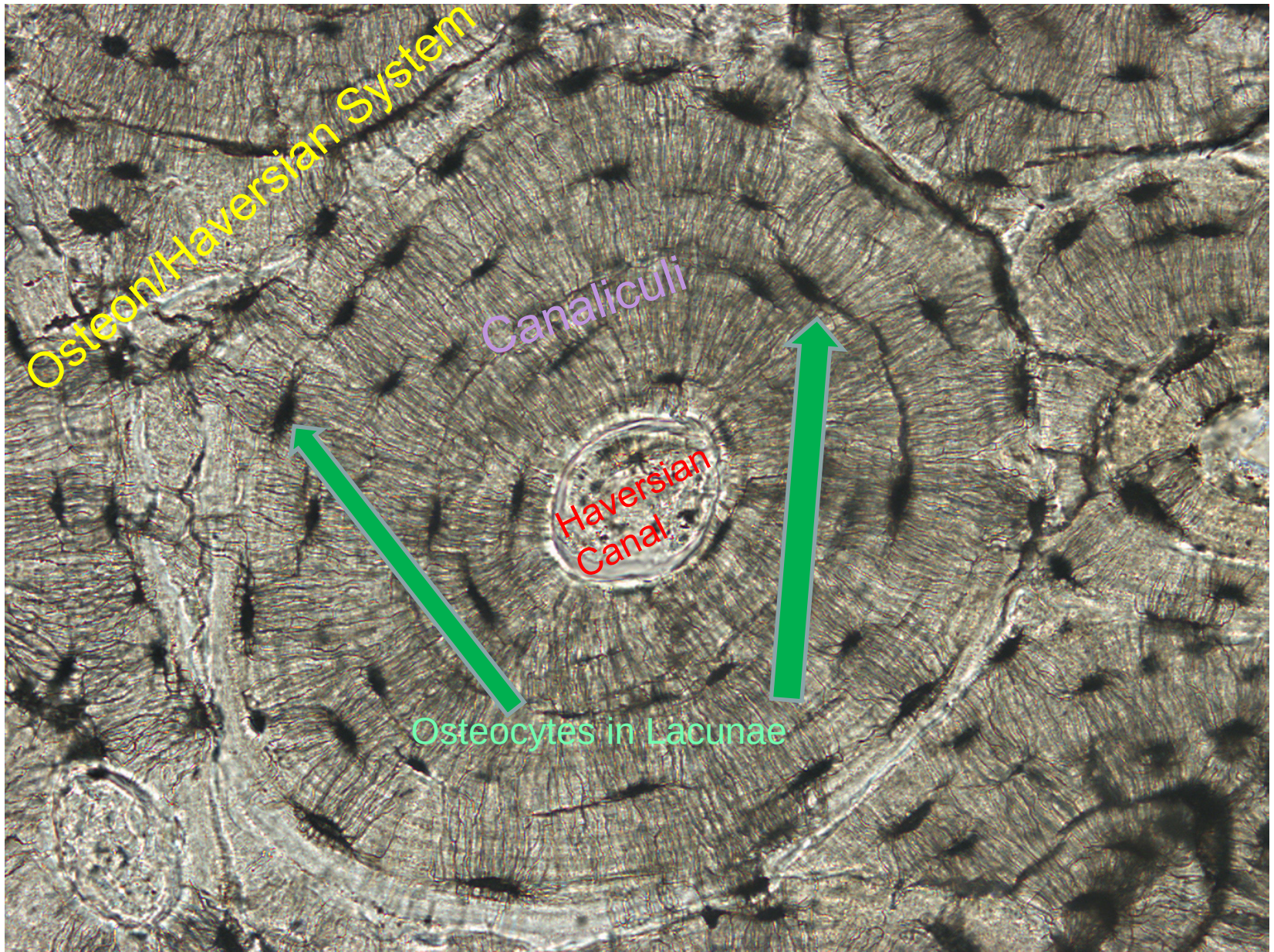


Compact bone

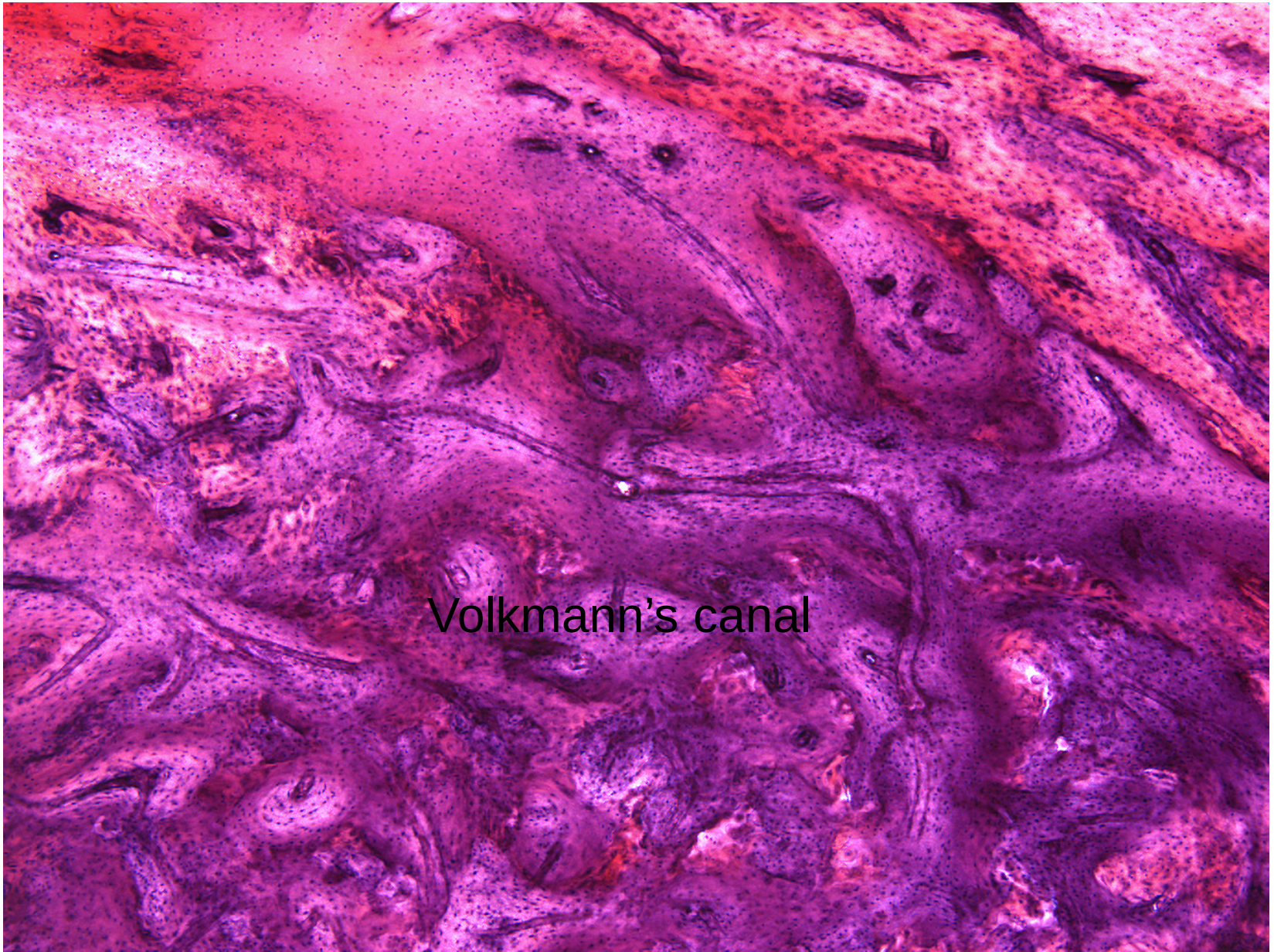
Gartner & Hiatt illustration depicting compact bone, osteons, Volkmann's and Haversian canals, and cancellous (spongy) bone



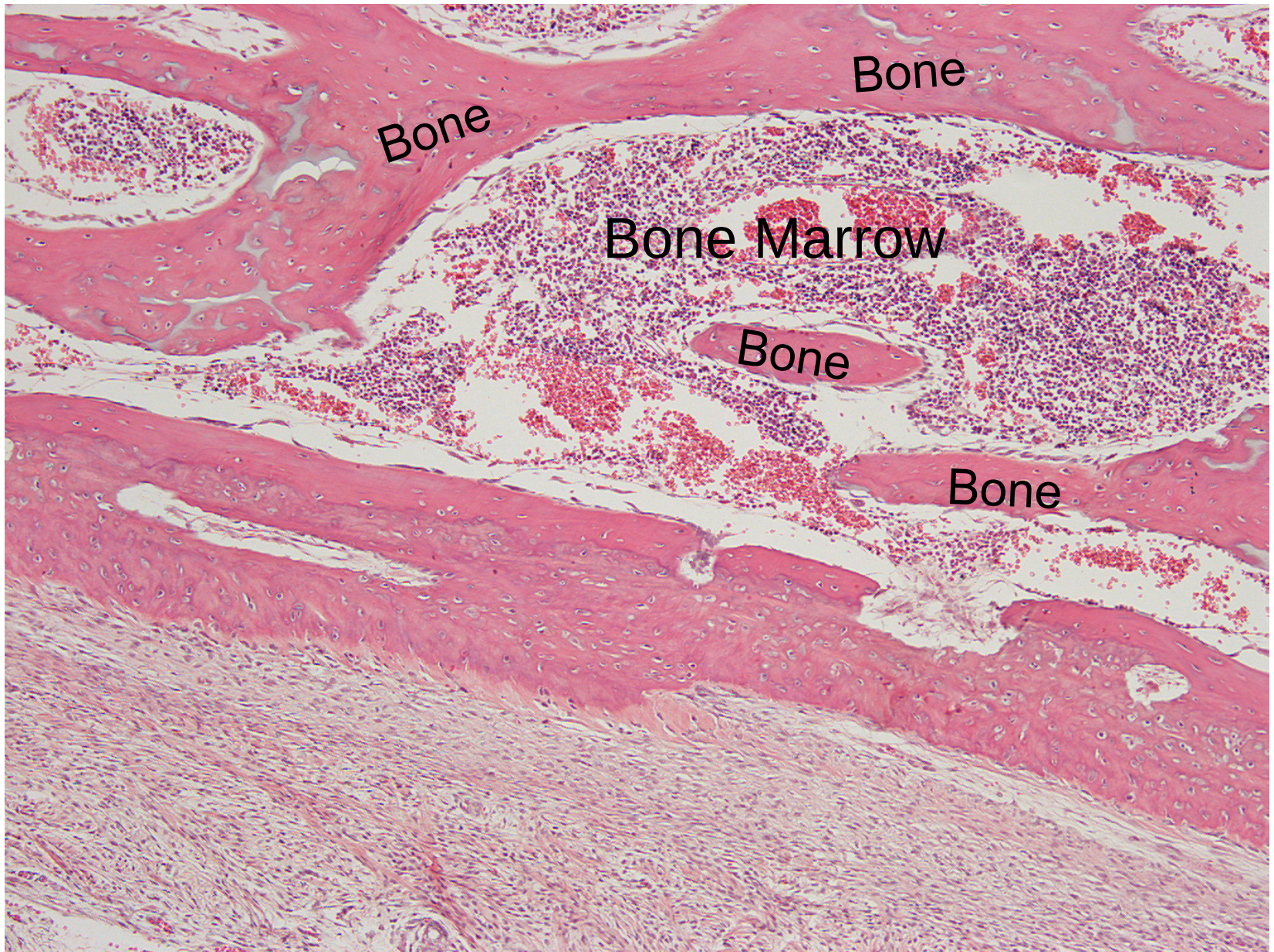
Compact bone - femur



Compact bone – femur (high magnification) Osteon



Inner ear bone



Developing bone (Trabecular, cancellous, spongy bone)

Growth of Bone (Osteogenesis)

- 2 methods for bone growth:
 1. Intramembranous = Mesenchyme differentiates into osteoblasts
 2. Endochondral = cartilage becomes bone
- There is no interstitial growth of bone (osteocytes do not divide)

Intramembranous Ossification

1. Mesenchyme differentiates into osteoblasts
2. Osteoblasts create a matrix (osteoid)
3. Osteoid calcifies and forms bone spicules
4. Spicules form patches of woven bone
5. Blood vessels invade area
6. Woven bone becomes mature bone – remodeling of osteoblasts and osteoclasts which form layers of compact bone that surrounds trabecular (spongy) bone

Endochondral Ossification

1. Mesenchyme differentiates into chondroblasts

Cartilage can then grow via appositional and interstitial growth

2. Chondrocytes hypertrophy

as perichondrium is vascularized and becomes periosteum and osteoprogenitor cells become osteoblasts

3. Osteoblasts create matrix

New matrix cuts blood supply to chondrocytes, causing them to die. Lacunae are left behind = framework for bone marrow

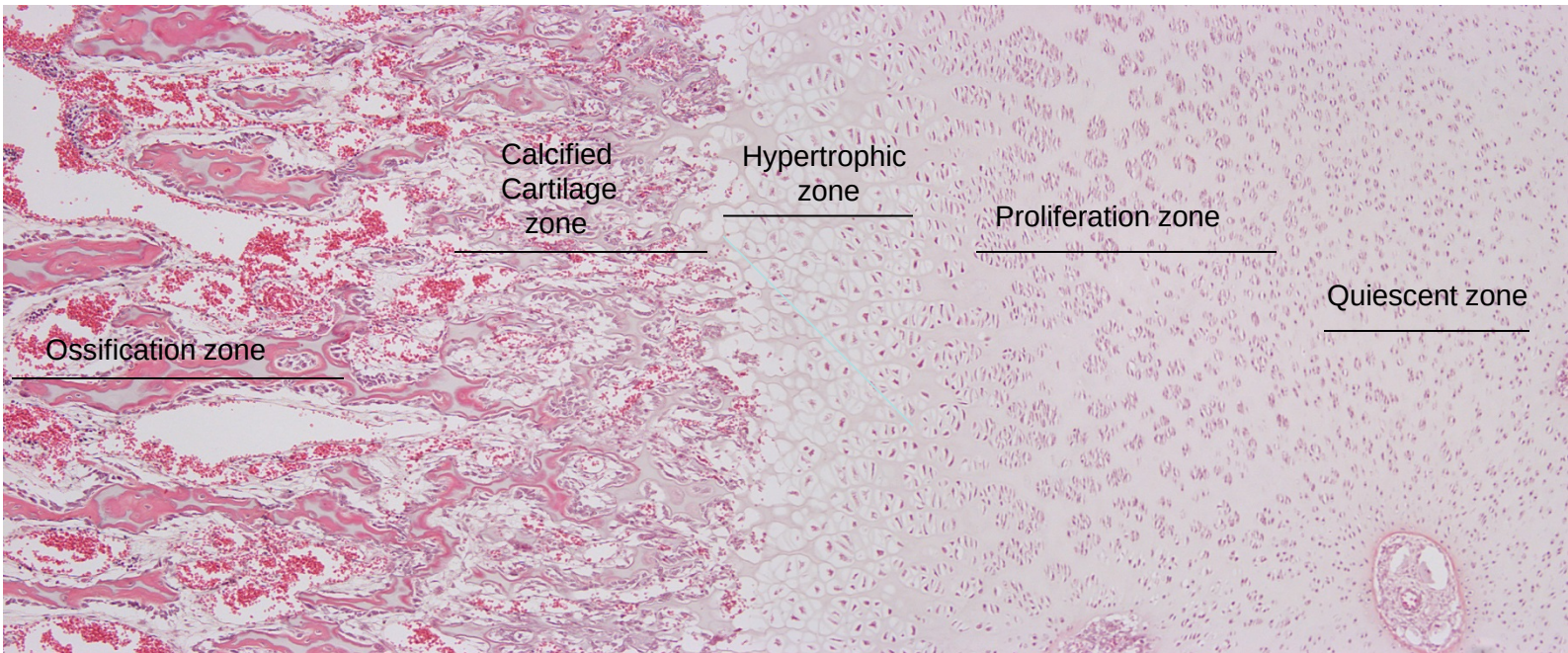
4. Osteoclasts etch holes in bone

Allows osteoprogenitor, hemopoietic cells and blood vessels to enter

5. Osteoprogenitor cells form osteoblasts which lay bone matrix over cartilage matrix

6. Bone grows thick and long

Endochondral Ossification



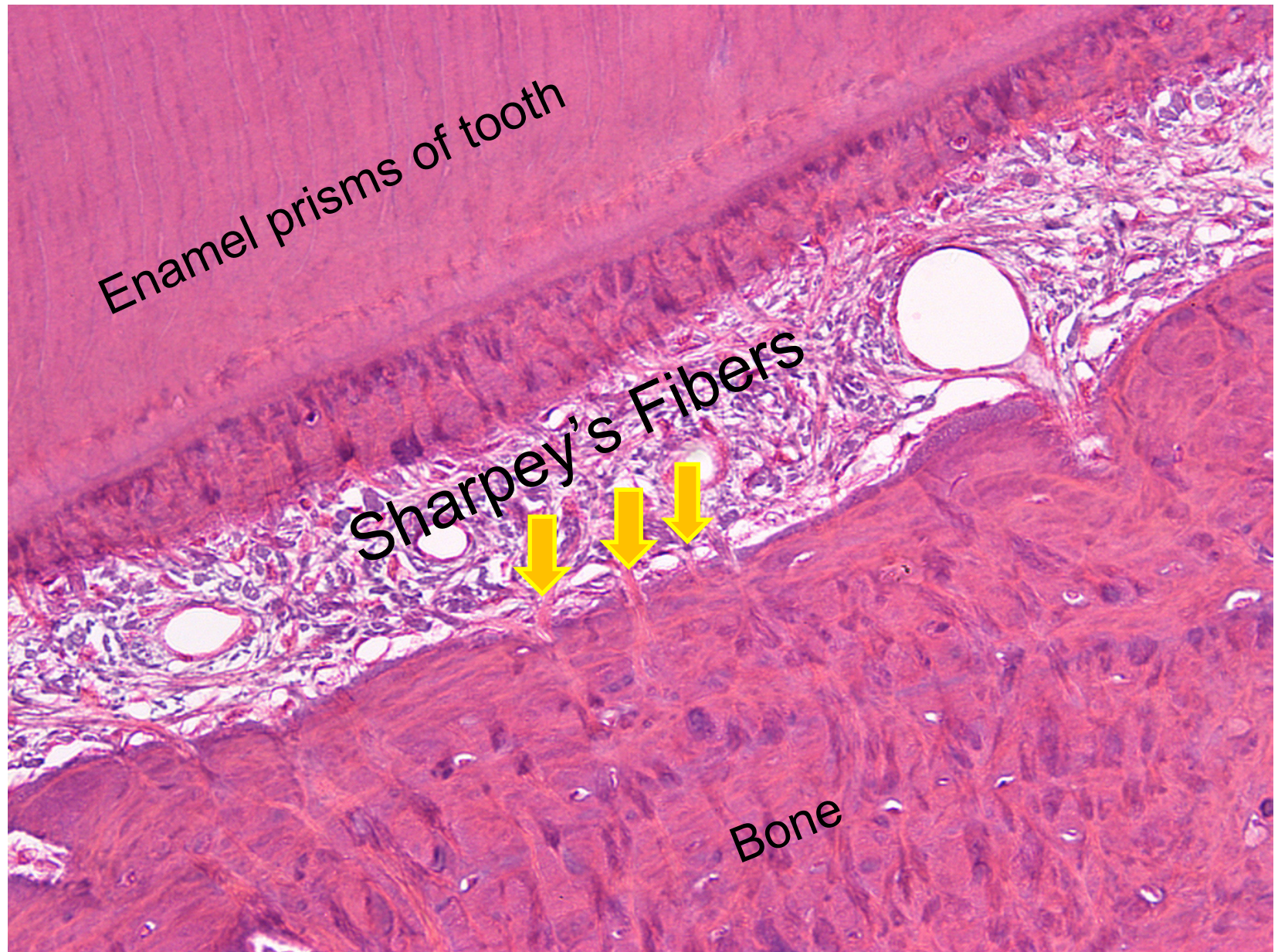
1. Reserve cartilage

2. Isogenous groups

3. Chondrocytes mature and hypertrophy

4. Chondrocytes die, matrix calcifies

5. Osteoblasts create matrix



Canine mandible